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SARDAR PATEL UNIVERSITY
External Examination (CBCS)
B. Sc. (CA & IT) / M. Sc. - Information Technology (Integrated)
II^{nd} - Semester (CBCS)
PS02CIIT03: Advanced 'C' Programming and Introduction to Data Structures
17^{th} April, Tuesday - 2018

Time : 10:00 am to 1:00 pm   Total Marks :70

Q-1 Select an appropriate option. 10

1. Which of the following is not a C memory allocation function?
(a) malloc() (b) realloc() (c) calloc() (d) alloc()

2. malloc() function sets the initial value in memory as _____.
(a) zero (b) NULL (c) garbage (d) none of these

3. Which operator is used with a pointer to access the value of the variable whose address is contained in the pointer?
(a) Address (&) (b) Assignment (=) (c) Indirection (*) (d) Selection (->)

4. f = fopen(filename, "r");
Referring to the code above, what is the proper definition for the variable f?
(a) FILE f; (b) FILE *f; (c) struct FILE f; (d) int f;

5. What are two predefined FILE pointers in C?
(a) stdout and stderr (b) console and error
(c) stdout and stdio (d) stdio and stderr

6. Which of the following allows a portion of memory to be shared by different types of data?
(a) Array (b) Structure (c) Union (d) File

7. Files are a ____________________type of Data Structure.
(a) Linear (b) Primitive (c) Non-Primitive (d) Non-Linear

8. Which of the following data structure store the homogeneous data elements?
(a) Arrays (b) Records (c) Pointers (d) None of these

9. A data structure that contains not only a data field but also contains pointer field is known as _________.
(a) Queue (b) Stack (c) Tree (d) Linked List

10. A data structure in which insertion and deletion of an elements occurs at both the end is known as _______.
(a) Stack (b) Queue (c) Priority Queue (d) Deque

(P. T. O.)

Q-2 Answer the following questions. (Attempt any TEN) 20

1. Give the concept of pointers to array.
2. Define: structure, member operator.
3. Differentiate: a pointer and a pointer variable.
4. Explain the Append mode with example.
5. What do you mean by file pointer?
6. Explain typedef in brief with suitable example.
7. What do you mean Linear Data Structure?
8. State various Applications of Stack.
9. Which are the main operations that can be performed on Data Structure?
10. Define: Queue and Deque.
11. What is a Circular Linked list?
12. What is a doubly Linked list?

Q-3

(a) Define pointer variable. How can we declare and initialize pointer variable? How can we access value of variable through pointer type variable? 5

(b) Explain array of structures using suitable examples. 5

OR

Q-3

(a) Write a note on Dynamic memory allocation. 5

(b) Explain pointer to structure using suitable example. 5

Q-4

(a) Explain fprintf() and fscanf() function with example. 5

(b) Describe the usage and limitation of function getc() and putc(). 5

OR

Q-4

(a) Explain the getw() and putw() function with example. 5

(b) What is union? Explain its definition, declaration and assigning values to members of union. 5

Q-5

(a) Write down advantages of data structure. 5

(b) Write an algorithm to delete an element from a Stack. 5

OR

Q-5

(a) Write a short note on non linear data structure. 5

(b) Write an algorithm to insert an element into a Stack. 5

Q-6 Write an algorithm to insert an element into and delete an element from a simple queue. 10

OR

Q-6 Write an algorithm to insert an element at the beginning of the singly linked list and at the end of singly linked list. 10

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